



Known

$$F = 400 \text{ lbf}$$

$$E = 10 \times 10^6 \text{ psi}$$

$$\delta_{\max} = 0.009 \text{ in} \quad d = 0.5 \text{ in}$$

$$A = \frac{\pi (0.5)^2}{4} = 0.19635 \text{ in}^2$$

bar length

$$\delta = \frac{FL}{AE} \rightarrow L = \frac{\delta AE}{F}$$

$$L = \frac{(0.009)(0.19635)(10 \times 10^6)}{400}$$

$$L = 44.2 \text{ in}$$